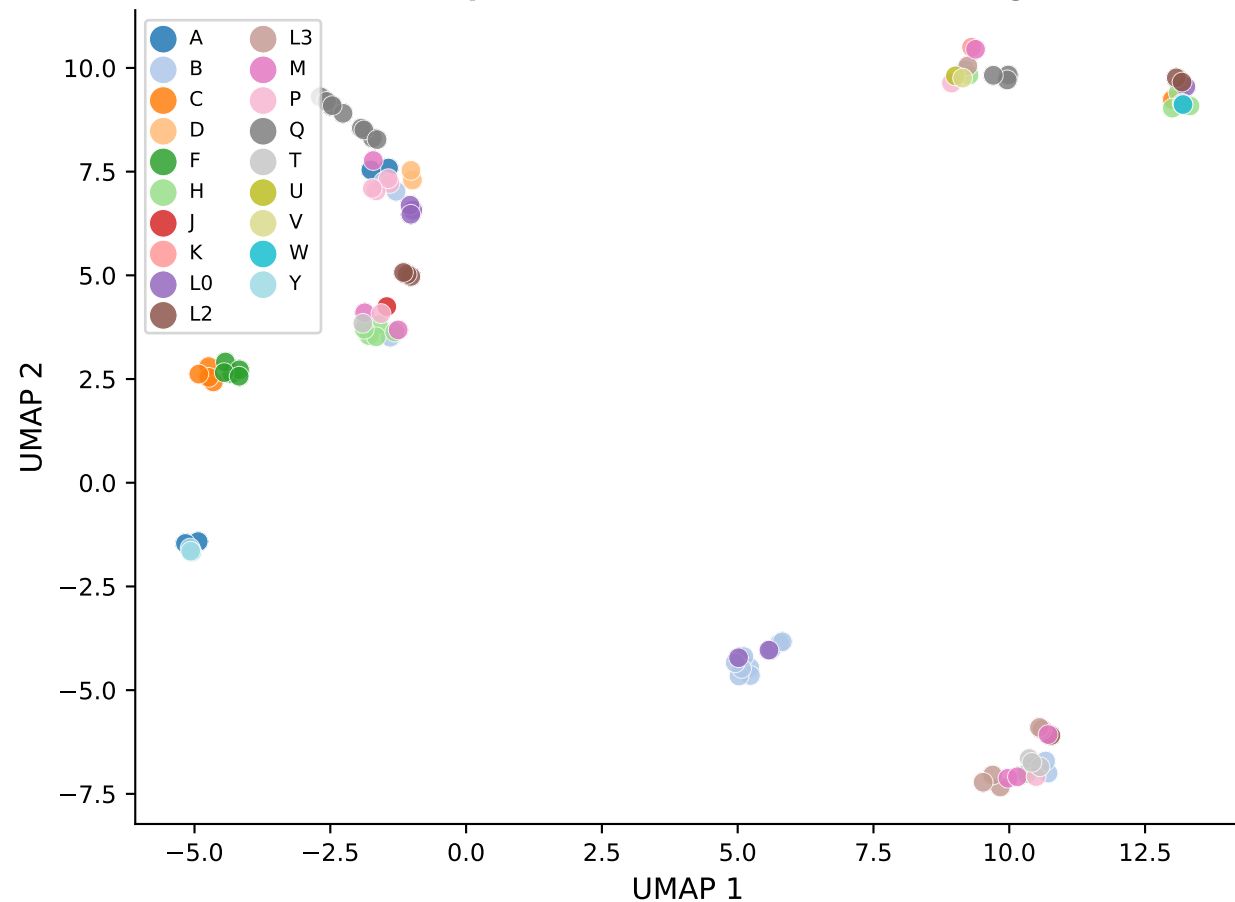


Figure 3: Haplogroup embedding structure and zero-shot classification accuracy

**(a) mtDNA-FM embeddings in UMAP space
(100 sequences, zero-shot, no fine-tuning)**



**(b) Zero-shot haplogroup verification
(8-class panel, zero-shot 3-NN)**

